

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA

Electronics and Communication Engineering Department

Data Structure and Algorithms (IIPC 210)

Mid Sem Exam-II

Max. Marks:15

Duration: 50 min.

Date:

All questions are compulsory.

Q1. You are given the classic N-Queens problem, but with the following additional constraints:

Place N queens on an $N \times N$ chessboard such that no two queens attack each other. Additionally:

- Some cells are blocked (represented by x) and cannot be used.
- One or more queens are pre-placed at fixed positions and cannot be moved.
- You must print all possible valid configurations that satisfy the above.

Q2. Find the Topological Ordering of the graph given in Fig. 1.

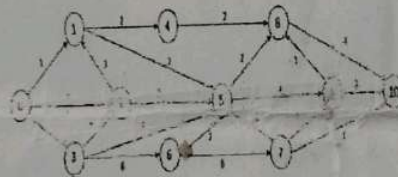


Fig. 1

Q3. Solve the following TSP which is represented as directed graph given in Fig. 2.

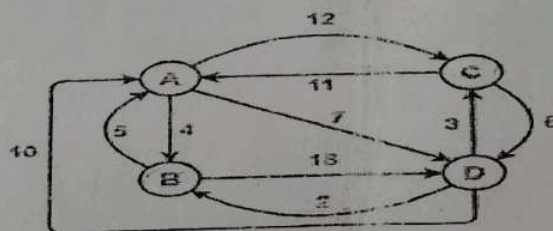


Fig. 2